INDIVIDUAL PROJECT REPORT

ON

“Arya College ClubSphere: Centralized Club and Event Management Platform”

INDIVIDUAL CONTRIBUTION REPORT — AUTHENTICATION & USER MANAGEMENT

Submitted in partial fulfillment of the requirement for the award of Degree of

Bachelor of Technology in Computer Science & Engineering

Submitted to:

Rajasthan Technical University, Kota (Raj.)

Submitted by

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Enrollment No.: 24EARCS032

Individual responsibility: Authentication & User Management

PBL Team ID: PBL2627-CSE-A-024

Under the guidance of

Er. Ram Babu Buri

Department of Computer Science & Engineering

Session: 2026–27

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CERTIFICATE

This is to certify that the individual project report entitled “Arya College ClubSphere — Authentication & User Management” has been prepared by Avantika Mishra (24EARCS032), a student of Bachelor of Technology in Computer Science & Engineering at Arya College of Engineering & I.T., Jaipur, affiliated with Rajasthan Technical University, Kota, during the academic session 2026–27.

The report records the student’s individual contribution to the group project Arya College ClubSphere (PBL2627-CSE-A-024), carried out under the guidance of Er. Ram Babu Buri. The contribution described here forms part of a four-member team project; the work of the other team members is acknowledged and is not claimed in this report.

Role	Name	Signature
Project Guide	Er. Ram Babu Buri Dept. of CSE, ACEIT, Jaipur	
Head of Department	Dr. Akhil Pandey Dept. of CSE, ACEIT, Jaipur	
Student	Avantika Mishra (24EARCS032)	

Date: _____ Place: Jaipur

DECLARATION

I, Avantika Mishra (24EARCS032), a student of B.Tech. Computer Science & Engineering at Arya College of Engineering & I.T., Jaipur, hereby declare that this individual report describes my own contribution to the group project Arya College ClubSphere, carried out during the session 2026–27 under the guidance of Er. Ram Babu Buri.

The technical content of this report was prepared from the project source code, database schema and module documentation. Features built by teammates are stated as such. The report does not claim sole authorship of ClubSphere or measured production performance.

To the best of my knowledge, this report has not been submitted to any other university or institution for the award of any degree or diploma.

Avantika Mishra

Enrollment No.: 24EARCS032

B.Tech. (CSE), Session 2026–27

Signature: _____ Date: _____

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to my project guide, Er. Ram Babu Buri, for his guidance throughout the planning and development of ClubSphere. His emphasis on clear requirements and honest reporting of status shaped the way I designed and documented the Authentication & User Management module.

I am thankful to Dr. Akhil Pandey, Head of the Department of Computer Science & Engineering, Arya College of Engineering & I.T., for the academic environment, and to the faculty of the department for their support.

I am grateful to my teammates — Avanshi Kothari, Akansha Sharma and Aditi Maheshwari. My module depends on a shared MySQL schema, AuthenticationFilter and JSP chrome. Working with Git taught me integration and communication within a team.

Finally, I thank my family and friends for their encouragement and patience during the project.

Avantika Mishra

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ABSTRACT

This individual report documents my contribution to Arya College ClubSphere (PBL2627-CSE-A-024): the Authentication and User Management module. I implemented login and logout, BCrypt password handling, session-fixation mitigation, AuthenticationFilter role checks, and the administrative user directory (create user, change role, toggle status with a self-lockout guard). Every other module depends on a trusted currentUser session attribute. Verification included PasswordUtil unit tests and manual role-access checks.

Keywords: BCrypt, HTTP session, RBAC, Jakarta Servlet, AuthenticationFilter

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CHAPTER 1 — INTRODUCTION

1.1 Background

ClubSphere is the group PBL project (PBL2627-CSE-A-024) that centralises clubs, events, QR registration, tasks and notifications for Arya College of Engineering & I.T. The team report describes the whole system. This individual report documents only my contribution: **Authentication & User Management**.

1.2 Purpose of this Report

This report separates my work from that of my teammates, explains the design of my module, how it was tested, and what remains incomplete.

1.3 Problem Addressed by My Module

Without a single login and role gate, club, event and dashboard pages cannot be protected. Students must not open /admin by typing the URL. Passwords must not be stored in plaintext.

1.4 Objectives of My Contribution

- Authenticate with username or college email and BCrypt passwords.
- Create a fresh HTTP session on login and invalidate it on logout.
- Enforce ADMIN, CLUB_HEAD, CLUB_MEMBER and STUDENT URL prefixes.
- Let administrators manage users without deactivating their own session.
- Keep the login redirect parameter limited to internal paths.

CHAPTER 2 — PROJECT OVERVIEW AND INDIVIDUAL RESPONSIBILITIES

2.1 Overall Project Context

ClubSphere is developed by four students and hosted at github.com/Avantikamishra862/Arya_Clubsphere. It is a Java 17 / JSP / Servlet / MySQL application deployed as a Tomcat WAR.

2.2 Module Ownership

Member	Enrollment	Module
Avantika Mishra	24EARCS032	Authentication & User Management
Avanshi Kothari	24EARCS031	Club Management
Akansha Sharma	24EARCS012	Event Management & QR Code Registration
Aditi Maheshwari	24EARCS007	Dashboards, Tasks & Notifications

My owned area is highlighted in the implementation chapter. Shared files (schema.sql, AuthenticationFilter, navbar.jsp, pom.xml) were coordinated with the team.

2.3 In-scope and Out-of-scope

In scope: LoginServlet, LogoutServlet, AuthenticationFilter, UserService, UserDAO, PasswordUtil, login.jsp, admin users views.

Out of scope: I do not claim club CRUD, event QR generation, task assignment or dashboard widgets except where they consume currentUser.

CHAPTER 3 — TECHNOLOGY STACK

I used the shared ClubSphere stack. My module specifically depends on the items below.

Layer	Technology	How I used it
Security	jBCrypt 0.4	Hash and verify passwords (cost 10)
Web	Jakarta Servlet 6 / Tomcat 10.1	Login/logout servlets and @WebFilter
Session	HttpSession + web.xml	HttpOnly cookie, 60-minute timeout
Data	users table, UserDAO	Parameterized SQL for identity
Test	JUnit 5 PasswordUtilTest	Hash/verify and null handling

CHAPTER 4 — REQUIREMENTS OF MY MODULE

4.1 Users

All four roles authenticate through this module. Only ADMIN may create users and change roles.

4.2 Functional Requirements

ID	Requirement
FR-A1	Login with username or email
FR-A2	BCrypt storage and verification
FR-A3	Role dashboard routing
FR-A4	Filter protection of role prefixes
FR-A5	Admin user create / role / status
FR-A6	No self-deactivation of the logged-in admin

4.3 Non-functional requirements

Fail closed on missing session. Do not leak stack traces. Use parameterized SQL. Do not commit real database passwords.

CHAPTER 5 — DESIGN AND IMPLEMENTATION

5.1 Request flow

Browser → AuthenticationFilter → LoginServlet → UserService → UserDAO → users table. On success, session is renewed and the user is redirected.

5.2 Principal classes and views

Layer	Items
Filter	AuthenticationFilter (/*)
Controllers	LoginServlet (/login), LogoutServlet (/logout), AdminServlet user paths
Service / DAO	UserService, UserDAO, PasswordUtil
Views	login.jsp, WEB-INF/views/admin/users.jsp

5.3 Implementation notes

Public whitelist includes /, /login, /clubs, /events, /qr/ and static assets. Deep links preserve a safe return URL. Login.jsp includes evaluator autofill for seeded demo accounts. Legacy seed hashes can be upgraded on successful login.

UML figures for the whole system are in the team report and in docs/UML/Arya_College_ClubSphere_UML_Diagrams.pdf.

CHAPTER 6 — TESTING AND VERIFICATION

6.1 Strategy

PasswordUtilTest (JUnit) plus manual tests: valid/invalid login, logout, student hitting /admin, admin role change.

6.2 Representative checks

Check	Result / note
Hash and verify password123	Unit test passed
Wrong password	Login error, no session
Student opens /admin	Forbidden / redirect
Admin toggles own status	Blocked with error message

CHAPTER 7 — CHALLENGES AND LEARNING OUTCOMES

7.1 Challenges

Aligning Jakarta APIs with Tomcat 10; keeping the filter whitelist in sync with public club/event/QR pages; upgrading seed hashes without breaking demo logins.

7.2 Learning outcomes

I learned session-fixation defence, RBAC as a filter rather than a UI-only rule, and why secrets belong in environment variables.

CHAPTER 8 — CONCLUSION AND FUTURE SCOPE

8.1 Conclusion

The authentication module is the gate for ClubSphere. With sessions, RBAC and BCrypt in place, other modules can assume a trusted User object.

8.2 Future work in my module

Optional public self-registration, HTTPS-only Secure cookies in production, and more automated filter tests.

REFERENCES

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[2] Apache Tomcat 10.1 Documentation.

[3] MySQL 8.0 Reference Manual.

[4] BCrypt password hashing documentation.

[5] ZXing QR code library documentation.

[6] Arya College ClubSphere team project report, Dept. of CSE, ACEIT, Jaipur, Session 2026–27, PBL2627-CSE-A-024.

[7] https://github.com/Avantikamishra862/Arya_Clubsphere